

Wazuh-Based Security Monitoring and Threat Hunting SOC Lab

Introduction

This project focused on building a controlled Security Operations Center (SOC) lab using Wazuh for centralized log monitoring, endpoint visibility, alert correlation, and threat hunting. The environment included a Wazuh Manager on Ubuntu Server, a Windows 11 endpoint, and a Kali Linux system used for controlled attack simulation and monitored Linux telemetry.

The lab validated the following controlled security-monitoring scenarios:

- File Integrity Monitoring (FIM)
- Registry integrity monitoring
- User account activity monitoring
- Firewall-based scan detection
- Threat hunting using MITRE ATT&CK mapping
- Multi-endpoint log collection and analysis

The goal was to understand how a SIEM collects endpoint telemetry, correlates events, and supports investigation of suspicious activity across multiple systems.

Scope and Environment

All testing was performed in a controlled private lab using systems owned or administered for the exercise. No production or third-party systems were targeted.

Lab Environment and Network Configuration

System	Purpose / Role	IP Address	Wazuh Version
Ubuntu Server	Wazuh Manager / SIEM	192.168.1.74	4.14.5
Windows 11	Monitored endpoint	192.168.1.70	4.7.5
Kali Linux	Attacker + Wazuh Agent	192.168.1.71	4.14.5

Tools and Technologies Used

Tool	Purpose
Wazuh	SIEM monitoring and alerting
Kali Linux	Attack simulation
Nmap	Reconnaissance simulation
Windows Defender Firewall	Firewall telemetry
Wazuh Agents	Endpoint log collection
File Integrity Monitoring (FIM)	File and registry monitoring
MITRE ATT&CK Framework	Threat classification
Windows Event Logs	Endpoint security telemetry
Wazuh Indexer	Alert storage and search
Wazuh Dashboard	Investigation and threat hunting interface

Project Architecture

Flow	Source / Activity	Traffic / Telemetry	Destination / Analysis
1	Kali Linux (Attacker)	Nmap / controlled network traffic	Windows 11 endpoint
2	Windows 11 Wazuh Agent	Security, FIM and Firewall telemetry	Ubuntu Wazuh Manager
3	Kali Linux Wazuh Agent	Linux endpoint security telemetry	Ubuntu Wazuh Manager
4	Wazuh Manager	Decoded alerts, rules and correlation events	Wazuh Indexer
5	Security Analyst	Queries and threat hunting	Wazuh Dashboard

Attack traffic is generated from Kali against the Windows endpoint; endpoint and agent telemetry is forwarded to the Wazuh Manager for correlation and analysis, then surfaced through the Dashboard and Threat Hunting views.

Project Implementation

Detection Methodology

Attack or endpoint activity was generated in the lab, collected by the Wazuh agent, processed and correlated by the Wazuh Manager, indexed for search, and reviewed through the Wazuh Dashboard. Findings were then classified using event context, rule severity, and MITRE ATT&CK mapping.

1. Wazuh Deployment

Ubuntu Wazuh Manager Setup

Installed and configured:

- Wazuh Manager
- Wazuh Dashboard
- Wazuh Indexer

Verified:

- Wazuh services running successfully
- Dashboard accessibility

Final Wazuh stack version: 4.14.5.

- Agent connectivity

2. Endpoint Integration

Windows 11 Agent

Installed Wazuh Agent on Windows endpoint.

Configured:

- Manager IP address
- Agent communication
- Log forwarding

Verified:

- Agent status active in Wazuh dashboard
- Windows logs successfully ingested

Kali Linux Agent

Installed and configured the Wazuh Agent on Kali Linux as both a monitored Linux endpoint and the attack-simulation host.

Resolved:

- Version mismatch issues
- Agent authentication problems
- Manager connectivity configuration

Verified:

- Kali endpoint successfully connected and became active in the Wazuh Dashboard.

Agent Registration and Connectivity Validation

Registered each endpoint with the Wazuh Manager, configured the manager address, completed agent authentication, started the Wazuh agent service, and verified active status in the Wazuh Dashboard. Kali agent communication was validated on TCP/1514.

The Kali agent initially reported a version-compatibility error because its 4.14.5 agent was newer than the manager. The Wazuh Manager stack was upgraded to 4.14.5, after which the Kali agent connected successfully.

Proof of Successful Agent Connectivity

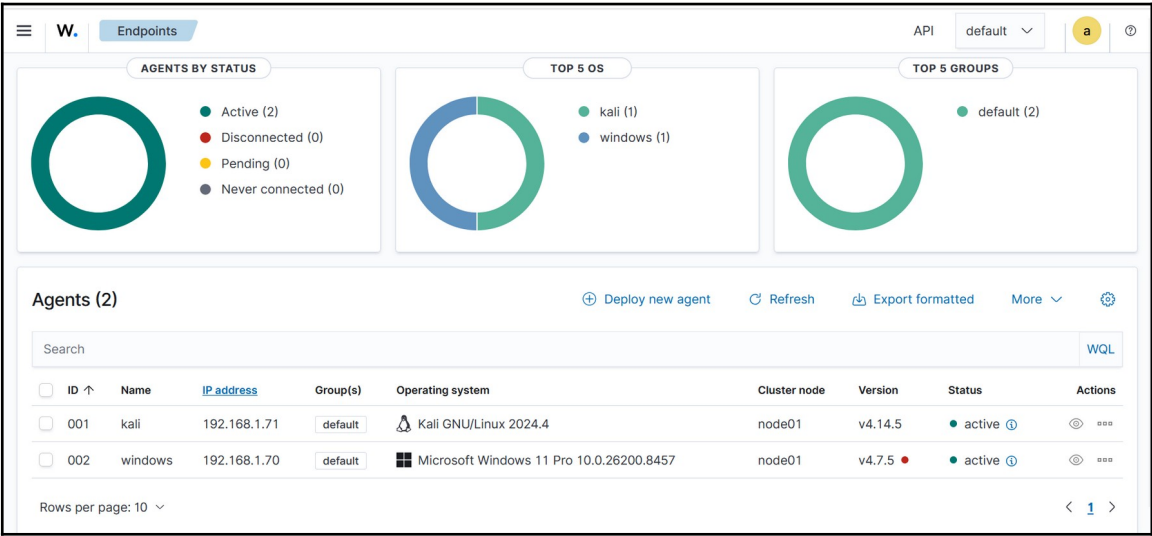


Figure 1: Active Wazuh agents connected to centralized manager.

Log Collection and Telemetry Sources

Telemetry Source	Endpoint / Component	Purpose
Windows Security Events	Windows 11	Authentication and account activity
File Integrity Monitoring (Syscheck)	Windows 11	File and registry integrity changes

Windows Firewall Log	Windows 11	Network connection and dropped-packet telemetry
Wazuh Manager / Indexer	Ubuntu Server	Rule processing, correlation, storage and search

3. File Integrity Monitoring (FIM)

Objective

Monitor selected Windows directories and registry activity for integrity changes that may require investigation.

Configuration

Modified Windows agent configuration:

```
<directories check_all="yes">C:\Users\Public</directories>
```

Reduced monitoring interval:

```
<frequency>60</frequency>
```

Restarted Wazuh agent after configuration changes.

Attack Simulation

Performed a controlled FIM test by creating and modifying a file inside the monitored directory:

```
echo hacked > C:\Users\Public\test.txt
```

```
echo modified >> C:\Users\Public\test.txt
```

Detection Results

Wazuh generated the following integrity events during the lab:

- Integrity checksum changed alerts
- Registry key integrity alerts
- Registry value integrity alerts

Detected Rule IDs:

- Rule 550 — Integrity checksum changed
- Rule 594 — Registry key integrity checksum changed
- Rule 750 — Registry value integrity checksum changed

Severity Analysis

Rules 594 and 750 were observed at Wazuh level 5, while Rule 550 was observed at level 7. These were treated as FIM integrity events requiring investigation; an integrity change alone was not treated as proof of malicious activity.

Proof of File Integrity Monitoring Alerts

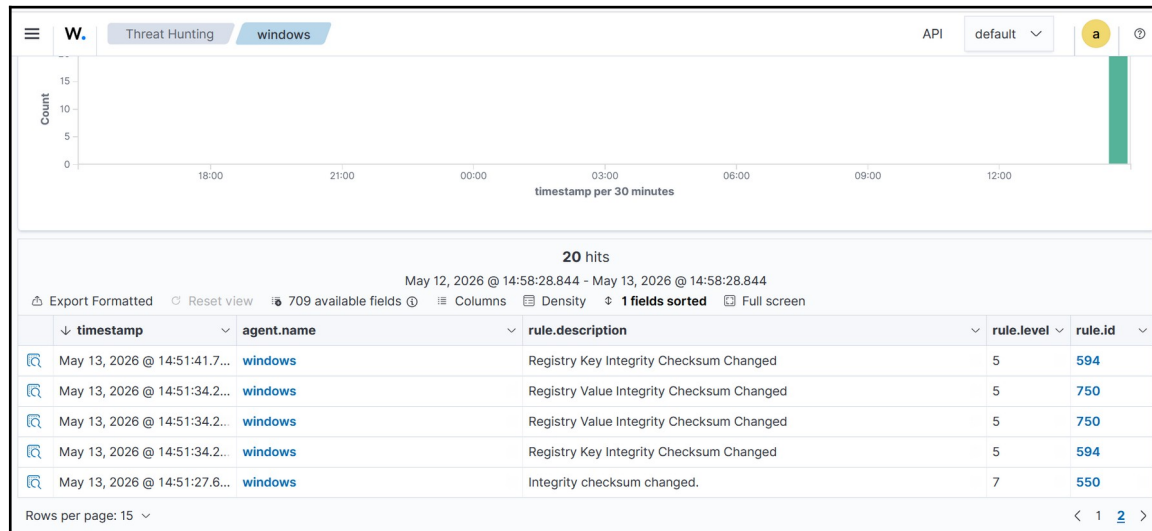


Figure 2: Wazuh detecting registry and integrity checksum changes during FIM testing.

Detection Interpretation

The lab generated file-integrity and registry-integrity events. Because Windows and installed applications can legitimately modify registry data, these alerts were treated as investigation leads rather than automatically classified as persistence or compromise.

Impact

Demonstrated the ability to identify:

- unauthorized file modifications
- registry integrity changes that may warrant persistence investigation
- suspicious endpoint activity

4. Firewall Telemetry Integration

Objective

Integrate Windows Firewall logs into Wazuh to detect suspicious network activity.

Configuration Steps

Enabled:

- dropped packet logging
- successful connection logging

Firewall log location:

C:\Windows\System32\LogFiles\Firewall\pfirewall.log

Integrated firewall logs into Wazuh:

```
<localfile>
  <location>C:\Windows\System32\LogFiles\Firewall\pfirewall.log</location>
  <log_format>syslog</log_format>
</localfile>
```

Restarted Wazuh agent after configuration.

5. Network Scan Detection

Objective

Detect reconnaissance activity generated from the Kali Linux attack-simulation host against the Windows endpoint.

Attack Simulation

Performed a controlled Nmap SYN scan from Kali Linux against the Windows endpoint:
nmap -sS 192.168.1.70

Detection Results

Wazuh correlated repeated firewall-drop events and generated a higher-level detection:

Rule ID	Description
4151	Multiple Firewall drop events from same source

Proof of Reconnaissance Detection

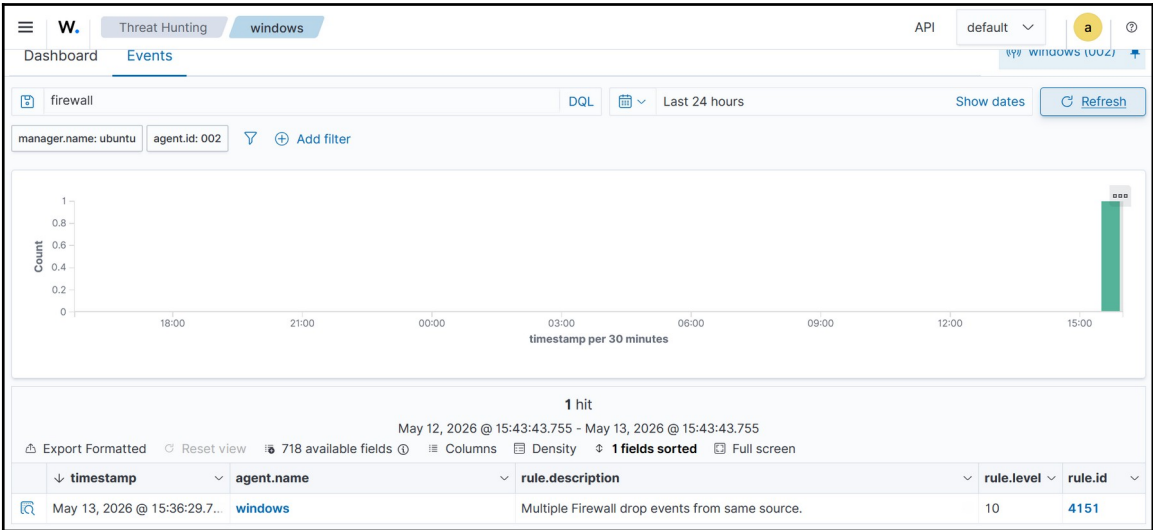


Figure 3: Wazuh correlated firewall drop events generated during Nmap scan activity from Kali Linux attacker system.

Alert Investigation Workflow

Stage	Activity	Outcome
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1. Attack	Kali performed Nmap SYN scan against Windows endpoint.	Reconnaissance traffic generated.
2. Telemetry	Windows Firewall recorded dropped packets.	Firewall telemetry available to Wazuh.
3. Detection	Wazuh correlated repeated firewall drops.	Rule 4151 generated.
4. Investigation	Reviewed source endpoint, timestamp, and event context.	Activity identified as network scanning.
5. Mapping	Mapped reconnaissance behavior to MITRE ATT&CK T1046.	Technique classified.
6. Response	Recommended review/blocking of the source and reduction of unnecessary exposed services.	Containment guidance documented.

Network Detection Severity and Correlation

Rule 4151 was observed at Wazuh level 10 in the lab. The rule correlated repeated firewall-drop events from the same source into a higher-level alert rather than generating a separate alert for every packet, reducing repetitive alert volume.

During investigation, the underlying firewall events can be reviewed to validate source IP, destination, protocol, ports, and timestamps before deciding on response actions.

MITRE ATT&CK Mapping

Technique ID	Description
T1046	Network Service Discovery

Use Case	Primary Telemetry	Observed Evidence	MITRE / Rule	Analyst Action
FIM / Registry	Wazuh syscheck	Rules 550, 594, 750	FIM integrity events	Validate change and affected path
Nmap Scan	Windows Firewall log	Rule 4151, level 10	T1046 / Network Service Discovery	Investigate source, ports and exposure
Account Creation	Windows Security log	Event 4720; T1098 observed	T1098 / Account Manipulation	Validate user, initiator and authorization
Failed Logon	Windows Security log	Event 4625; Rule 60122	MITRE mapping as investigation lead	Check source, frequency and account status

Proof of MITRE ATT&CK Event Mapping

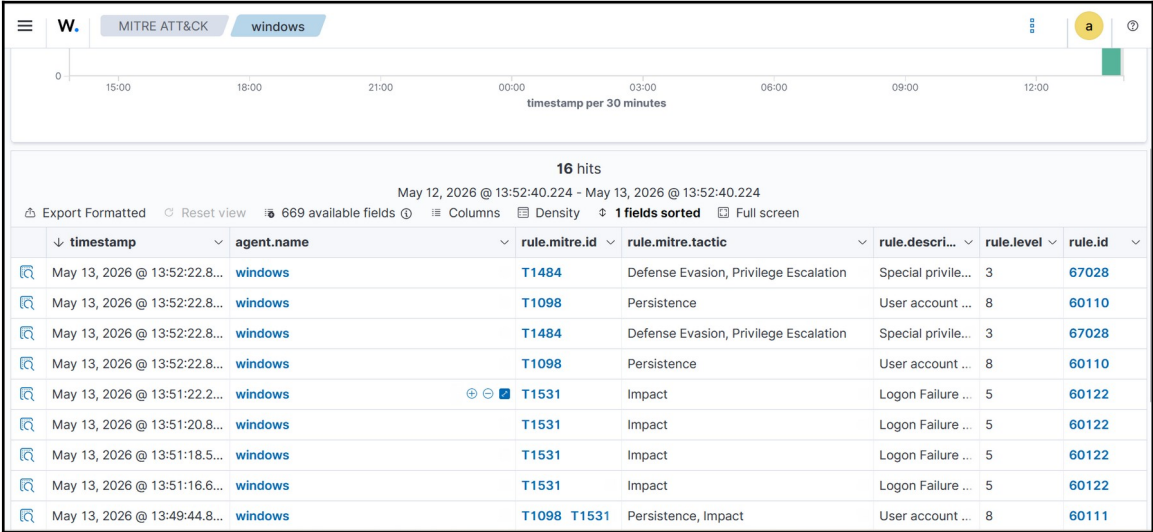


Figure 4: Wazuh Threat Hunting view showing Windows account, logon-failure, and MITRE ATT&CK mapped events.

Impact

Demonstrated:

- reconnaissance detection
- firewall telemetry analysis
- SIEM event correlation
- threat hunting capability

6. Windows Account and Authentication Monitoring

Objective

Detect local account creation and repeated authentication failures on the Windows endpoint using Wazuh-collected Windows Security events.

Account Creation Test

A controlled local account was created on the Windows endpoint using:
net user attacker Pass123! /add

Detection Result

Windows Event ID 4720 was generated for the controlled local-account creation. Wazuh displayed account-related alerts, including an observed MITRE ATT&CK mapping to T1098 (Account Manipulation). The mapping was treated as supporting context; the primary evidence was the underlying Windows account-creation event.

Authentication Failure Test

Repeated incorrect Windows login attempts were generated in the lab to produce Windows Security Event ID 4625 authentication-failure telemetry.

Detection Result

Wazuh generated logon-failure events, including Rule 60122 in the observed dashboard view. The event was validated against the underlying Windows Security telemetry and treated as an investigation lead rather than proof of a malicious attack by itself.

Response Considerations

For a production alert, an analyst would validate the affected account, source context, timing, and related events before disabling an unauthorized account, forcing a credential reset, or escalating the incident. No automated containment was executed in this lab.

7. Threat Hunting and MITRE ATT&CK Analysis

Used Wazuh Threat Hunting module to:

- investigate alerts
- analyze correlated events
- identify attack patterns
- map detections to MITRE ATT&CK techniques

Observed tactics included:

- Persistence
- Defense Evasion
- Privilege Escalation
- Impact

Key Achievements

- Built a functional multi-endpoint SOC lab
- Integrated Windows and Kali Linux endpoints
- Configured centralized log monitoring
- Implemented File Integrity Monitoring (FIM)
- Integrated Windows Firewall telemetry
- Detected reconnaissance activity using Wazuh correlation rules
- Performed threat hunting using MITRE ATT&CK framework
- Simulated controlled attack scenarios safely within the lab environment

Challenges Faced

Challenge	Resolution
Agent version mismatch	Upgraded Wazuh Manager stack to 4.14.5 and revalidated agent compatibility.
Kali agent connection issues	Corrected manager address/communication settings, authenticated the agent, and verified TCP/1514 connectivity.
Missing firewall telemetry	Enabled Windows Firewall logging and onboarded pfirewall.log through a Wazuh localfile configuration.

XML configuration errors	Corrected the localfile XML syntax and restarted the Wazuh agent.
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Capability	Status	Evidence / Notes
Multi-endpoint Wazuh lab	Completed	Ubuntu Manager, Windows 11 agent and Kali agent active
Centralized log monitoring	Completed	Windows Security, FIM and Firewall telemetry
File Integrity Monitoring	Completed	Syscheck configuration and integrity alerts
Firewall telemetry / scan detection	Completed	pfirewall.log + Rule 4151 during Nmap scan
Threat hunting / MITRE mapping	Completed	Wazuh Threat Hunting views and ATT&CK mapping
VirusTotal enrichment	Not implemented	Future enhancement
Sysmon endpoint telemetry	Not implemented	Future enhancement

Limitations

This version of the lab focused on detection, investigation, and threat classification. Automated containment, VirusTotal enrichment, and Sysmon-based endpoint telemetry were not implemented and remain future enhancements. These items are intentionally excluded from the completed-project claims.

The FIM scan interval was reduced to 60 seconds for laboratory testing; production deployments should tune scan frequency according to workload and monitoring requirements.

Conclusion

This project demonstrated how Wazuh can be used as a centralized SIEM platform for monitoring endpoint activity, detecting suspicious behavior, and performing threat hunting across multiple systems.

The lab successfully simulated attacker activity from Kali Linux and validated Wazuh's ability to detect:

- file modifications
- registry changes
- firewall events
- reconnaissance activity

The project provided practical exposure to SIEM operations, endpoint monitoring, detection engineering, log analysis, SOC workflows, and MITRE ATT&CK-based threat hunting.